

Track 2E Offset Predictability Feasibility

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report consumes the completed `Track 2D` mean-offset artifacts and asks whether the vertical curve offset is predictable enough from causal condition information to justify an offset-aware next branch.

This is an analysis-only feasibility diagnostic. It does not train models, alter the dataset, update registries, or use future TE samples as model inputs.

- Run Instance: `2026-06-03-13-28-54__track2e_offset_predictability_feasibility`
- Track 2D Source: `output\validation_checks\track2d_mean_offset_full_matrix_audit\2026-06-03-10-54-10__track2d_mean_offset_full_matrix_audit`
- Output Directory: `output\validation_checks\track2e_offset_predictability_feasibility\2026-06-03-13-28-54__track2e_offset_predictability_feasibility`
- Candidate Count: `111`

Method

- candidate raw, centered-shape, and offset metrics are imported from `Track 2D` ;
- candidate offsets are summarized by causal groups: direction, speed, torque, oil temperature, and their direction-aware combinations;
- exact full-condition groups such as speed plus torque plus oil temperature are intentionally excluded from the recommendation ranking because they can collapse to one evaluated curve and overstate deployable offset predictability;
- the corrected `MAE` is an upper-bound diagnostic approximation:
centered `MAE` plus the remaining absolute offset after subtracting a group mean offset;
- the correction baseline is not a production model and is not a valid registry promotion rule;
- `Fw` , `Bw` , and `global` are interpreted as parallel branches.

Intervention Counts

Surface	Intervention	Candidate Count
Bw	multi_head_shape_offset	19
Bw	not_offset_first	11
Bw	posthoc_offset_baseline	10
Bw	sequential_offset_model	1
Fw	multi_head_shape_offset	44
Fw	not_offset_first	8
Fw	sequential_offset_model	1
global	multi_head_shape_offset	12
global	not_offset_first	4
global	sequential_offset_model	1

Surface Recommendations

Rank	Candidate	Surface	Interv.	Raw MAE	Corr. MAE	Gain [%]	Explain [%]	Best Group
2	rcim_retuned_XGBM19_Bw	Bw	seq	0.010679	0.005572	47.8	72.6	dir_torque
3	LGBM19_Fw	Fw	seq	0.006812	0.003684	45.9	61.3	dir_torque
1	harmonic_regression_global	global	seq	0.018160	0.004604	74.6	84.2	dir_torque

Track 2E Feasibility Ranking

Rank	Candidate	Surface	Interv.	Raw MAE	Corr. MAE	Gain [%]	Explain [%]	Best Group
1	harmonic_regression_global	global	seq	0.018160	0.004604	74.6	84.2	dir_torque
2	rcim_retuned_XGBM19_Bw	Bw	seq	0.010679	0.005572	47.8	72.6	dir_torque
3	LGBM19_Fw	Fw	seq	0.006812	0.003684	45.9	61.3	dir_torque
4	XGBM19_Bw	Bw	post	0.007991	0.003925	50.9	64.7	dir_torque
5	GBM19_Bw	Bw	post	0.005180	0.002811	45.7	61.3	dir_torque
6	RF19_Bw	Bw	post	0.005392	0.003075	43.0	58.7	dir_torque
7	HGBM19_Bw	Bw	post	0.006619	0.003785	42.8	58.2	dir_torque
8	track1_best_Bw	Bw	post	0.005027	0.002917	42.0	57.9	dir_torque
9	LGBM19_Bw	Bw	post	0.005037	0.002947	41.5	57.9	dir_torque
10	ERT19_Bw	Bw	post	0.005258	0.003248	38.2	53.7	dir_torque
11	DT19_Bw	Bw	post	0.005226	0.003355	35.8	50.5	dir_torque
12	ET19_Bw	Bw	post	0.006273	0.004289	31.6	49.7	dir_torque
13	rcim_retuned_LGBM19_Bw	Bw	post	0.008105	0.005684	29.9	60.3	dir_torque
14	ELM19_Bw	Bw	head	0.010071	0.008031	20.3	37.9	dir_torque
15	XGBM19_Fw	Fw	head	0.004190	0.003489	16.7	37.5	dir_torque
16	GBM19_Fw	Fw	head	0.002779	0.002323	16.4	44.0	dir_torque
17	ELM19_Fw	Fw	head	0.007281	0.006370	12.5	30.3	dir_torque
18	rcim_retuned_ELM19_Fw	Fw	head	0.007182	0.006367	11.4	29.0	dir_torque
19	SVM19_Bw	Bw	head	0.004822	0.004372	9.3	33.6	dir_torque
20	rcim_retuned_ELM19_Bw	Bw	head	0.008917	0.008354	6.3	32.2	dir_torque

Largest Conservative Offset-Correction Gains

Rank	Candidate	Surface	Interv.	Raw MAE	Corr. MAE	Gain [%]	Explain [%]	Best Group
1	harmonic_regression_global	global	seq	0.018160	0.004604	74.6	84.2	dir_torque
4	XGBM19_Bw	Bw	post	0.007991	0.003925	50.9	64.7	dir_torque
2	rcim_retuned_XGBM19_Bw	Bw	seq	0.010679	0.005572	47.8	72.6	dir_torque
3	LGBM19_Fw	Fw	seq	0.006812	0.003684	45.9	61.3	dir_torque
5	GBM19_Bw	Bw	post	0.005180	0.002811	45.7	61.3	dir_torque
6	RF19_Bw	Bw	post	0.005392	0.003075	43.0	58.7	dir_torque
7	HGBM19_Bw	Bw	post	0.006619	0.003785	42.8	58.2	dir_torque
8	track1_best_Bw	Bw	post	0.005027	0.002917	42.0	57.9	dir_torque
9	LGBM19_Bw	Bw	post	0.005037	0.002947	41.5	57.9	dir_torque
10	ERT19_Bw	Bw	post	0.005258	0.003248	38.2	53.7	dir_torque
11	DT19_Bw	Bw	post	0.005226	0.003355	35.8	50.5	dir_torque
12	ET19_Bw	Bw	post	0.006273	0.004289	31.6	49.7	dir_torque
13	rcim_retuned_LGBM19_Bw	Bw	post	0.008105	0.005684	29.9	60.3	dir_torque
14	ELM19_Bw	Bw	head	0.010071	0.008031	20.3	37.9	dir_torque
15	XGBM19_Fw	Fw	head	0.004190	0.003489	16.7	37.5	dir_torque

Decision Interpretation

`sequential_offset_model` means the offset component appears large and condition-predictable enough to justify a future causal residual-offset probe. `posthoc_offset_baseline` means a simple causal aggregate calibration is worth keeping as a benchmark before training a second model.

`multi_head_shape_offset` means the next model should split centered waveform shape from offset / low-frequency behavior. `loss_reweighting` means the next safer step is a raw plus centered-shape plus offset loss.

`not_offset_first` means amplitude, phase, or centered shape should stay ahead of offset correction.

Runtime Input Boundary

The causal grouping baselines use only direction and operating condition metadata already present in the Track 2 payload. The analysis never gives a model the future TE curve. Any learned offset model still requires a later technical document and campaign plan before it can become a training branch.

Machine-Readable Artifacts

- `output\validation_checks\track2e_offset_predictability_feasibility\2026-06-03-13-28-54_track2e_offset_predictability_feasibility\track2e_candidate_feasibility_summary.csv`
- `output\validation_checks\track2e_offset_predictability_feasibility\2026-06-03-13-28-54_track2e_offset_predictability_feasibility\track2e_surface_intervention_recommendation.csv`
- `output\validation_checks\track2e_offset_predictability_feasibility\2026-06-03-13-28-54_track2e_offset_predictability_feasibility\track2e_condition_offset_stability.csv`
- `output\validation_checks\track2e_offset_predictability_feasibility\2026-06-03-13-28-54_track2e_offset_predictability_feasibility\track2e_offset_predictability_feasibility_summary.yaml`